COUNTRY PROFILES

Nuclear Power in Spain

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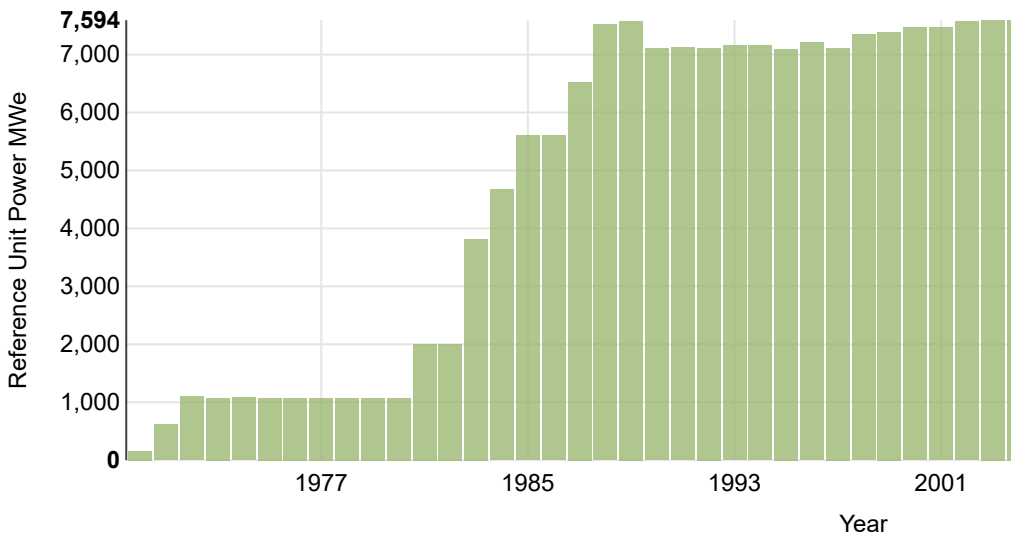
- Spain has seven nuclear reactors generating about a fifth of its electricity.
- Its first commercial nuclear power reactor began operating in 1968.
- Under Spain's nuclear phase-out policy, its nuclear fleet is scheduled to shut down by 2035. However, in 2025 parliamentary and industry initiatives called for this timetable to be reconsidered.

Privacy - Terms

The first reactor is planned to be taken offline in 2027.

7	0	3
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
7,123 MWe	0 MWe	1,070 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2024): 288 TWh

Generation mix: wind 62.1 TWh (22%); nuclear 54.5 TWh (19%); solar 53.7 TWh (19%); natural gas 71.6 TWh (18%); hydro 40.8 TWh (14%); oil 9.1 TWh (3%); biofuels & waste 7.0 TWh (2%); coal 3.5 TWh.

Import/export balance: 10.2 TWh net import (14.4 TWh imports; 24.6 TWh exports)

Total consumption: 230 TWh

Per capita consumption: c. 4700 kWh in 2024

Source: International Energy Agency and The World Bank. Data for year 2024.

At the end of 2024, installed capacity was 129 GWe, of which nuclear accounted for 7.1 GWe. As Spain is essentially separate from the EU grid – a small amount of power can be traded with France – electricity self-sufficiency is a major policy consideration.

Nuclear power industry

Reactors operating in Spain

Reactor Name	Model	Reactor Type	Net Capacity (MWe)	Construction Start	First Grid Connection
Trillo 1	PWR 3 loops	PWR	1,003	1979-08	1988-05
Almaraz 1	W (3-loop)	PWR	1,011	1973-07	1981-05
Almaraz 2	W (3-loop)	PWR	1,006	1973-07	1983-10
Asco 2	W (3-loop)	PWR	997	1975-03	1985-10
Asco 1	W (3-loop)	PWR	995	1974-05	1983-08
Cofrentes	BWR-6	BWR	1,064	1975-09	1984-10
Vandellos 2	W (3-loop)	PWR	1,047	1980-12	1987-12

In 1964 construction started on the first of three nuclear power reactors – Jose Cabrera, Zorita, a small pressurised water reactor. Two years later construction of Santa María de Garoña, a medium-sized boiling water reactor, was started, followed two years later by Vandellos 1, a medium-sized gas-cooled reactor similar to the UK's Magnox units. This first generation of Spanish units – all turnkey projects – gave practical experience with three different designs, and led to a focus largely on PWR types in the 1970s.

In 1972 ENUSA (Empresa Nacional del Uranio, SA, now now ENUSA Industrias Avanzadas SA), a state-owned company, was set up to take over all of the nuclear front-end activities.

In the early 1970s construction was started on a second generation of seven reactors, five of which were completed. These involved local engineering companies Empresarios Agrupados and INITEC and the state-owned manufacturer ENSA (Equipos Nucleares SA).

In the early 1980s, construction of a third generation of five units was started, but following a 1984 moratorium following the election of a socialist government, only two were completed – Trillo 1 and Vandellos 2. In 1994 the moratorium was confirmed and the units under construction and immediately planned were abandoned. These comprised two 1000 MWe Westinghouse reactors at Limoniz near Bilbao in the Basque region, which were ready to run, two 975 MWe GE reactors at Valdacaballeros in the Badajoz region, which were 70% and 60% complete, and Trillo 2, a planned German PWR. A total of €5.7 billion compensation was paid to Iberdrola and Endesa from 1994 through to 2015. The plants concerned were dismantled and the land sold. As well as these plants, plans for units at Deva and Tudela on the north coast, at Lugo and Sayago on the Portuguese border, and a 6000 MWe plant at Ispaster in the Basque region were abandoned.

In February 2011 parliament removed a legal provision limiting nuclear plant operating lifetimes to 40 years. The conservative government elected in November 2011 removed the 1984 moratorium, and early in 2012 an industry report recommended in principle 20-year operating lifetime extensions.

The Socialist government to 2011 had come to power on an anti-nuclear platform, but apart from opposing the licence renewal for the Santa María de Garoña plant, an early BWR-3 model, it became increasingly positive about nuclear power. In 2011 the responsible minister said that nuclear plants are "essential for the supply of electricity in Spain" and that almost all nuclear power units "will be open, operating and even repowering" until 2021.

Also, he said: "Nuclear energy will be useful as a source of electricity for cars," which the government was promoting, hoping then to have one million electric vehicles on the road by 2014. However, that government remained opposed to new nuclear plant construction. The Socialist government's anti-nuclear policy was never translated into legislation. The November 2011 election brought about a change of government which revisited the decision to close Garoña, potentially allowing operation to 2019 (see below).

Nuclear plant ownership and operation is mostly by the Spanish-based but now international utility Endesa SA (originally Empresa Nacional de Electricidad SA) and Iberdrola. Endesa is 70.71% owned by Italy's Enel. Endesa and Iberdrola have a joint venture operating company: Asociacion Nuclear Asco-Vandellos (ANAV) which covers 40% of the country's nuclear capacity, in Catalonia. Another joint operating company is Centrales Nucleares Almaraz-Trillo (CNAT), covering the central and west capacity. Centrales Nucleares del Norte (Nuclenor) owns and operated the Garoña plant in the northern province of Burgos. Nuclenor is also owned by Iberdrola and Endesa (50% each), and pioneered nuclear generation in Spain.

In January 2015 Gas Natural Fenosa (which took over Union Fenosa in 2009) said it would move its nuclear assets to its Gas Natural Generacion subsidiary.

In December 2023 the government announced its intention to close all nuclear power plants in the country, with the first reactor planned to be taken offline in 2027. The total cost of decommissioning is expected to be around €20.2 billion.

In February 2025 the Spanish Congress approved a non-binding proposal calling on the government to reverse the nuclear phase-out and to extend the operating lifetimes of the existing reactors in line with CSN safety guidance and economic criteria. Later that month companies representing the Spanish nuclear industry signed a manifesto urging that the 2019 phase-out agreement be renegotiated in the light of energy security, competitiveness and decarbonization goals.

In October 2025 the board of Centrales Nucleares Almaraz-Trillo (CNAT) decided to request a three-year extension of the operating licences for Almaraz 1&2 so that the units could operate until mid-2030 instead of closing in 2027. In November 2025 the Ministry for the Ecological Transition and the Demographic Challenge forwarded this application to the CSN. The ministry will decide on continued operation once it has CSN’s advice.

Power uprates

Spain is notable for power plant uprates and had a programme to add 810 MWe (11%) to its nuclear capacity through upgrading its reactors by up to 13%. For instance, the Almaraz nuclear plant was boosted by more than 5% at a cost of $50 million.

Cofrentes was uprated 2% in 1988, another 2.2% in 1998, 5.6% in 2002 and 1.9% in 2003, taking it to 112% of original capacity.

Licence renewal and taxes

Licence renewal for the **Santa María de Garoña** plant came up for review in 2009, and in June the Nuclear Safety Council (CSN) recommended that a 10-year extension be granted, to 2019. The CSN said that plant owner and operator Nuclenor had implemented a comprehensive work programme to keep the 40-year old reactor fully serviceable, having spent some €155 million on it. The Socialist government, with a policy then of closing down Spanish nuclear plants as early as possible, granted only a four-year licence extension, to July 2013. In January 2012 the new conservative government referred the matter back to CSN with a view to revoking the 2009 decision and allowing operation to 2019. CSN again approved this and in July 2012 the government removed the 2013 operational limit, subject to Nuclenor renewing the licence.

However, Nuclenor delayed its application until new government rules and taxes were specified, since €480 million in investment would be required to take the plant to mid-2019. Pending government plans involved heavy taxes* – 7% – on electricity generation, amounting to some 1 ¢/kWh for nuclear, plus a tax of €2190 per kilogram on used nuclear fuel discharged – about €315 million per year across the industry. In addition, utilities must make provision in their accounts for partially used fuel still in the reactor – this would amount to over €500 million per year. Having missed the September deadline to apply for licence renewal from July 2013, Garoña was shut down and promptly defuelled in December 2012, rather than paying taxes on the plant of €153 million in 2013 and €374 million to 2019, required under the energy reform bill which was passed in December. It was declared officially shut down in July 2013.

* The higher taxes on power generation are to address a €24 billion energy tariff deficit after more than a decade of selling electricity at

regulated rates which do not cover costs, with high subsidies on renewables – some €9 billion per year.

In May 2013 Nuclenor approached the government to seek a resolution of the impasse, which might allow the reactor to restart. The CSN said it supported the appeal, and the Ministry should allow Nuclenor to submit a request for a one-year extension of its operating licence before June. However it would need to complete some safety modifications before loading fuel. The Ministry responded in mid-June by confirming the mid-2013 shutdown, and saying it was “based solely on economic considerations”. In July 2014 Nuclenor was fined €18.4 million by the financial regulator for shutting down the plant earlier than originally authorized. The regulator said that Nuclenor committed "a very serious violation" of the country's laws governing the electricity sector by reducing production capacity or supply of electricity without permission.

In February 2014, the cabinet approved a royal decree that allows recently-shutdown plants to apply within 12 months for their operating licences to be renewed, if the shutdown was not for safety reasons. This opened the possibility of restarting Garoña if the company could secure a suitable deal. In May 2014 Nuclenor applied to the Ministry of Industry & Energy for a renewal of its operating licence to 2031, enough to justify spending €120 million on upgrading and also payment of the taxes. At the end of July 2014 the CSN voted in favour of issuing a technical instruction to Nuclenor on documentation and additional requirements for the Garoña operating licence renewal, and at the end of September 2014 the company submitted a detailed work plan for meeting these substantial requirements.

Nuclenor announced in March 2015 that inspections conducted in November and December 2014 indicated no manufacturing defects in the reactor pressure vessel and that it was in a good condition, enabling the unit to operate safely when restarted. The reactor remained closed while engineering work on CSN requirements proceeded, and in November 2015 over 400 people were working on the plant. In February 2017 CSN reported that it accepted the Nuclenor proposal for continued operation of the plant, subject to completion of works proposed by Nuclenor, and would recommend accordingly to the Ministry of Energy. Nuclenor said it would "analyse all aspects of the report, together with the rest of the applicable conditions, to decide on the future of the plant." Iberdrola was reported to be doubtful about the economic justification for upgrading the plant. In August 2017 the energy minister announced that the government would not approve the

renewal of Garoña's operating licence, but would issue a decommissioning permit.

In June 2016 the Tobalina Valley City Council granted a works permit for an independent spent fuel storage facility (ISFSI) for Garoña.

Early in 2019 a draft national energy and climate plan required by EU rules was agreed between the government, nuclear plant operators and the decommissioning and nuclear waste agency Enresa, setting a timetable for a nuclear power phaseout by 2035.

Licence renewal for **Almaraz 1&2** came up for review in 2010, and in April the CSN recommended that a 10-year extension be granted, to 2020. It said it had verified that the plant owners – Iberdola, Endesa and Union Fenosa – had kept commitments made at the last extension in 2000, and added that its decision was based on a thorough review of the plant's condition. The Ministry of Industry, Tourism & Trade then approved the CSN recommendation. In January 2011 the government approved 70 MWe uprates for both reactors, with 68 MWe for unit 1 being imminent, the engineering work having been already done. In March 2019 the owners announced that they would request licence renewal to November 2027 and 2028 for units 1&2 respectively. Licence renewal was granted in May 2020.

In June 2010 the CSN recommended a 10-year licence extension for **Vandellos 2**, and this was approved by the Ministry in July. In March 2019 ANAV (Endesa 72%) applied for a further 10-year licence extension to take the plant to July 2030. This was approved in July 2020. Its proposed closure under the draft energy and climate plan is in 2034.

In February 2011 the CSN recommended a 10-year extension for **Cofrentes**, and in July 2011 it recommended the same for **Asco 1&2**. These were agreed in March and September respectively. In February 2021 the CSN renewed Cofrentes' licence to November 2030. Licence renewals for both Asco 1&2 were granted in October 2021, to 2030 and 2031 respectively.

In October 2014 the CSN granted a 10-year licence renewal for Iberdrola's **Trillo**, subject to establishment of ageing management programmes. This was confirmed by the Ministry in November. A licence renewal is expected in March 2023. It is due to close in 2035.

In May 2016 Catalonia introduced a capacity tax on nuclear power plants in its territory, affecting three reactors – Asco 1&2 and

Vandellos 2. The rate is €44.14 per MWe per year.

A PricewaterhouseCoopers study in 2015 showed that the nuclear industry contributed a total of €2.78 billion to Spain's gross domestic product (GDP) in 2013, representing 0.27% of total GDP. The industry's direct contribution to GDP was €1.97 billion, or 0.19% of total GDP. The total tax contribution of the nuclear industry in 2013 was €1.14 billion, of which €781 million was taxes incurred on operation, which represented a cost for companies, and €360 million in taxes as a result of their business.

Energy Reform Bill

The Energy Reform Bill passed in December 2012 threatened the viability of renewable projects as subsidies were reduced, and risked increasing dependence on coal and also imported gas. Early in June 2013 the government announced a substantial reduction in renewables subsidies since they accounted for almost half of the €20 billion annual costs of the nation's electrical system, and capped retail tariffs were insufficient to pay for the open-ended cost of escalating renewable inputs. Subsidies for renewables totalled about €9 billion in 2012 and €9.3 billion in 2013. A deficit of €30 billion had built up since 2005, according to the energy and competition regulator CNMC.

The 2012 reforms had started to address this deficit, then in July the Ministry of Industry, Energy and Tourism introduced further ‘definitive reforms’ to reduce the deficit by €4.5 billion per year. These measures cost utilities €2.7 billion per year and consumers €400 million in 2013 and €900 million per year thereafter, while the government covered a further €2 billion in 2013 and €900 million per year thereafter of costs. Solar companies are expected to be worst affected, due to debt load estimated at €30 billion, and widespread financial distress was predicted by solar and wind industry groups. In 2000, the government had promised more than 20 years of large subsidies, and investment had proceeded on this basis. In May 2013 renewables received an average subsidy of €100/MWh. The reforms remove the feed-in-tariffs system and substitute a new regulated asset value-based system (or 'reasonable profitability' system).

At the start of 2014 the impact of the switch to capacity-based incentives was unclear. Enel Green Power said it expected to lose feed-in-tariffs (FITs) for one-third of its capacity installed before 2005, mostly for wind. All renewable sources now have to

take the pool price and there follows some uncertain assessment regarding “reasonable profitability”. In April 2014 CNMC said that proposed reductions in subsidies for renewables would cost producers some €1.7 billion in 2014 (wind €400 million, others €150-250 million each). The FIT modifications would determine the rate of return for existing renewable energy companies at 7.4% and for future ones at 7.5%, compared with more than 10% in the past. Iberdrola and Acciona were reviewing their business plans.

The new Renewable Energy Law was passed in June 2014 in order to reduce subsidies for renewables by €1.7 billion per year and control the tariff deficit which had reached €26 billion. The average payout from January to April 2014 was €90.70/MWh, total €3.39 billion. The decree applied to 39 GWe of renewable capacity and was designed to ensure an annual return of about 7.4% on invested capital over a project’s lifetime (20 years for wind plants). The capacity complement ranges up to €105/GWe for the newest wind plants, but tapers off to 2004 and older installations, mostly wind farms built before 2004, ceased to receive subsidies. The pool price calculation has caps and floors – for 2014 the caps were €52-56 and the floors €40-44, and these increased slightly to 2017.

Fuel cycle

There are no conversion or enrichment facilities in Spain, but ENUSA owns 11% of Eurodif, with a large diffusion enrichment plant at Marcoule in France that ceased operation in 2012. It also contracts for other conversion and enrichment services abroad.

ENUSA's Juzbado plant in Salamanca, commissioned in 1985, produces BWR and PWR fuel elements for Spain's reactors and also supplies other customers in Europe.

GNF ENUSA Nuclear Fuel S.A. (GENUSA) is jointly owned by Global Nuclear Fuel-Americas (a GE-led joint venture with Hitachi and Toshiba) as majority owner, and ENUSA. It was set up in 1989 and markets BWR fuel in Europe.

In 1991, ENUSA with Westinghouse Electric Corporation and British Nuclear Fuel Ltd (BNFL), created the European Fuel Group (EFG), with the purpose of a joint action in the European PWR nuclear fuel market.

Uranium mining

Uranium was discovered in Salamanca during the 1950s. Production commenced in 1974 at ENUSA's Fe mine (Mina Fe), which grew to become the largest uranium mine in the Iberian Peninsula. It produced over 4000 tU. The mine closed in 2000 due to low uranium prices, though minor output continued to 2002 from decommissioning, and the mining areas have since been restored. One large pit and three small ones are involved. At Mina Fe, the Elefante Plant was a bacterial heap leach facility which was replaced by the 800 tU/yr Quercus mill in 1993. The Quercus plant used a combination of heap leach and dynamic leach to 2003.

Australian-based Berkeley Energia Ltd owns uranium properties in the Salamanca province and also pursued a joint venture with ENUSA to develop the Salamanca State Reserves, using the Quercus mill (*Saelices el Chico*).* In July 2012 a new agreement between Berkeley and ENUSA gave Berkeley full rights to the Alameda and Villar deposits, in two state reserves. The extensively-drilled Alameda and Esperanza/Villar deposits are about 10km from Aguila and Quercus mill, and contiguous with other Berkeley leases against the Portugal border. Villar and Alameda North have 3660 tU inferred resources which are now part of Berkeley's Salamanca project. Almeda's 8130 tU is mostly indicated resources. A 2.5% royalty would apply. However, Berkeley waives its rights to mine in any state reserves where ENUSA has undertaken rehabilitation, or to use the Quercus mill.

* Apart from this Salamanca 1 project, in December 2008 Berkeley agreed with ENUSA to undertake an 18-month feasibility study on restarting uranium mining, focused on the Aguila and Alameda areas of the state reserves close to the Quercus mill. Berkeley had the right to acquire up to 90% of ENUSA's mining and exploration assets, including the Quercus mill which would then form part of its Salamanca Project. Following Berkeley's notification in January 2011 that it wished to proceed with mining the state reserves, ENUSA had until 29 February 2012 to form this new joint venture company to take forward the $228 million proposal. The new company was not incorporated by then and Berkeley took ENUSA to the Paris-based International Court of Arbitration of the International Chamber of Commerce. Berkeley claimed compensation for a total of over $200 million for the damages and losses caused by ENUSA's breach of the cooperation agreement. In April 2012 ENUSA said that it considered that the project was not viable, and therefore any JV company was irrelevant. The ENUSA state reserves in the Aguila Area include three significant deposits – Sageras, Majuelos and Palacios – two of which have been previously mined as Mina Fe. These have 8000 tU (JORC-

compliant), half of this as measured and indicated resources. Berkeley had raised A$ 55 million through a share issue and was to pay €20 million for the 90% equity in ENUSA assets if and when these were transferred into a new joint venture company which would develop the project. This JV would be 90% Berkeley and 10% ENUSA.

Berkeley's own Retortillo area, including some smaller deposits, is 35 km northeast of Alameda. While Berkeley attempted to progress the ENUSA JV project, in October 2011 the company turned its attention back to its own deposits where exploration had ceased in 2008, and it applied for a mining licence for these as Salamanca 1, using bacterial heap leaching rather than toll processing through the Quercus mill (as envisaged with the JV).

In April 2014 Berkeley was granted a 30-year mining licence for Retortillo. Possible satellite operations include initially Zona 7 (11,600 tU, 10 km from Retortillo), and in July 2015 CSN approval represented the first of three steps in authorizing the treatment plant as a radioactive facility in the name of Berkeley Minera Espana. In October 2015 the company had all approvals in hand for project infrastructure.

Initial onsite infrastructure work began in 2016, and work began on road realignment and a power line upgrade ahead of main construction.

In August 2017 the company arranged finance to bring the project into production through a $120 million investment agreement with Oman's sovereign wealth fund. It comprised $65 million as a convertible loan that will give the fund a 28% share of the company and fully finance development of the Retortillo deposit, and a $55 million option package for later stages of the project, which would allow the fund to purchase a further 9% share. Oman will also have the opportunity to become an offtake partner.

The municipality of Retortillo in 2020 issued Berkeley Energia a permit for construction works at the project, leaving the authorization for construction of the uranium concentrate plant as a radioactive facility (referred to as 'NSC II' authorization) as the only outstanding approval required for full construction to begin.

In November 2021, the Ministry for Ecological Transition and the Demographic Challenge (MITECO) rejected Berkeley's NSC II application following an unfavourable report issued by Spain's Nuclear Safety Council (*Consejo de Seguridad Nuclear*, CSN). Later the same month, Berkeley submitted a supplementary

improvement report and corresponding arguments addressing the issues raised by the CSN and requested a reassessment.

Berkeley said MITECO had rejected its NSC II application "without following legally established procedure"and submitted an appeal against MITECO's decision in December 2021. Berkeley said in July 2023 that it had received formal notification from MITECO that its appeal was rejected.

The Salamanca project has 59.8 million pounds U_3O_8 (23,000 tU) of measured and indicated resources as well as inferred resources of 29.6 million pounds of U_3O_8. According to a definitive feasibility study published by Berkeley in 2016, it would be capable of producing an average of 4.4 million pounds of uranium per year at $13.30 per pound over an initial ten-year period.

The 1600 tonnes of uranium used in Spain each year is imported. ENUSA has a 10% stake in COMINAK, which operates Niger's Akouta mine that ceased production in March 2021.

Waste management

ENRESA (Empresa Nacional de Residuos Radiactivos SA) was established in 1984 as a state-owned company to take over radioactive waste management and decommissioning of nuclear plants. It is now the only state-owned part of the nuclear fuel cycle in Spain.

It drew up a general plan for radioactive waste which was approved by parliament in 1999. It is based on nuclear power plant operating lifetimes of 40 years, and addresses the need to manage almost 200,000 cubic metres of low- and intermediate-level waste and 10,000 cubic metres of used fuel and other high-level waste.

ENRESA's low- and intermediate-level waste storage facility is at El Cabril, Cordoba. It has operated since 1961. In July 2016 Enresa was given approval by the Nuclear Safety Council to operate the new VLLW disposal facility there, which has a storage capacity of over 17,000 m^3. In February 2024 the government approved the environmental impact statement for the construction of a new southeast platform at El Cabril with 27 additional disposal cells for low- and intermediate-level radioactive waste, to be built in two phases. Twelve cells are planned to enter service around 2028, with the remaining 15 to be constructed later according to storage needs.

Since 1983 Spain's policy has been for an open fuel cycle, with no reprocessing. The 1999 plan for used fuel envisaged initial storage at each reactor for ten years. Some temporary storage for dry casks at Trillo up to 2010 was planned, and establishment of a longer-term centralised facility would follow. Meanwhile research would progress on deep geological disposal as well as transmutation, with a decision on disposal to be made after 2010. Granite, clay and salt formations were under consideration.

In mid-2006 Parliament approved ENRESA's 6th General Radioactive Waste Plan including a proposal to develop a temporary central nuclear waste storage facility by 2010, and the CSN approved its design, which was similar to the Habog facility near the Borssele power plant in the Netherlands. In December 2009 the government called for municipalities to volunteer to host this €700 million Almacen Temporal Centralizado (ATC) facility for high-level waste and used fuel. The government offered to pay up to €7.8 million annually once the facility is operational. It is designed to hold for 100 years 6700 tonnes of used fuel and 2600 m^3 of intermediate-level waste, plus 12 m^3 of high-level waste from reprocessing the Vandellos 1 fuel. The facility is to be built in three stages, each taking five years. Asco and Villar de Canas were two towns among eight that volunteered, attracted by the prospect of €700 million over 20 years and the annual direct payments, plus many jobs. A campaign of fearmongering was mounted by nuclear detractors to dissuade residents of the eight towns, and some regional governments were also opposed.

In September 2011 the Ministry for Industry announced its selection and rankings:

- Zarra (Valencia) 736 points.
- Asco (Tarragona) 732 points.
- Yebra (Guadalajara) 714 points.
- Villar de Canas (Cuenca) 692 points.

In December 2011 the Ministry announced that Villar de Canas had been selected, though only a 60-year storage period was mentioned. Pending construction, used fuel remains at individual power plants. Most used fuel is in storage pools. Trillo, Asco, Almaraz and Jose Cabrera also use dry storage, with other plants planning to construct similar interim storage solutions.

In July 2015 the CSN confirmed that Villar de Canas was suitable for the ATC and that ENRESA could begin preliminary site works while the regulatory process continued. The ATC facility would provide storage for some 12,816 cubic metres of waste for 60 years, by which time a repository for permanent disposal should be available. In 2018 the Ministry for the Ecological Transition halted the regulatory process.

As of December 2018, 15,838 irradiated fuel elements were stored at Spanish nuclear power plants, with 14,085 stored in pools, and 1753 in dry storage. Holtec has licensed and supplied its HI-STAR 100 dual-purpose metal casks to transport fuel from the individual power plants to the ATC central storage facility. The casks can also store used fuel.

Waste management and decommissioning is funded by an ENRESA levy of of €6.6/MWh on electricity sales. As of April 2019 its funds were reported as about €5.8 billion, against an estimate of €13 billion required by 2070 for decommissioning.

In July 2022 the first container holding used fuel from the Garoña plant was placed in the onsite interim dry storage facility. Forty four additional containers were to be manufactured for the remaining 2245 used fuel rods in the storage pool. In May 2025 the NSC approved an expansion of the onsite interim dry storage facility for used fuel at Garoña, together with an updated waste and used fuel management plan for the plant so that all fuel can be accommodated during decommissioning.

Decommissioning

In Spain, once a plant is shut down and a decommissioning permit is granted, control is transferred from owners and operators to ENRESA, which is responsible both for decommissioning and long-term management of radioactive waste.

Vandellos 1, a 480 MWe French UNGG gas-graphite reactor, was closed down in mid-1990 after 18 years' operation, due to a turbine fire which made the plant uneconomic to repair. In 2003 ENRESA concluded phase 2 of the reactor decommissioning and dismantling project, which allows much of the site to be released. The cost of the 63-month project was €93 million. After 30 years in Safestor, when activity levels have diminished by 95%, the remainder of the plant will be removed.

In April 2006 the 142 MWe Jose Cabrera (Zorita) PWR plant was closed after 38 years' operation. Rather than using Safstor, dismantling the plant is being undertaken over eight years from 2010 by Enresa – on schedule and within budget, the total cost is estimated at €150 million at 2016 prices. About 4% of the plant's constituent material will need to be disposed of as radioactive waste, the rest can be recycled, including 43 tonnes of internal components. Used fuel is stored onsite. The demolition of the last large building at the plant, the turbine building, was completed in June 2022.

In December 2012 the Santa Maria de Garoña plant, an early BWR-3 model of 446 MWe net, operating since 1971, was closed down after 41 years' operation. Nuclenor operated the plant on behalf of its owners, Iberdrola and Endesa. For four years there appeared to be some prospect of restarting the plant (see section above). Dismantling of the plant began in July 2023.

Shutdown Spanish reactors

	Type	Net MWe	First power	Shutdown	Years of operation
Vandellos 1	UNGG	480	1972	1990	18
Jose Cabrera (Zorita)	PWR	142	1968	2006	38
Santa Maria de Garona	BWR	446	1971	2012	41

Regulation, safety & non-proliferation

In 1980 the Consejo de Seguridad Nuclear (Nuclear Safety Council, CSN) was set up to take over both nuclear safety and radiological protection matters. The CSN was overhauled in 2007, following an incident in 2004 at Vandellos 2, and the scope for penalties increased.

In 2009 Endesa was fined €15.4 million over a radioactive release incident during a refuelling operation at Asco 1 in 2007. There were six charges of breaching safety rules. The incident was rated 2 on the INES scale.

Licensing is under a 1964 law (amended) and 1999 regulations by the Economic Ministry, advised by CSN and Ministry of Environment.

Civil liability for nuclear damage is covered under international conventions to which Spain is a party – the IAEA Vienna Convention and the OECD Paris and Brussels Conventions. Operators need to cover €150 million.

Non-proliferation

Spain is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1967 and in 1985 it came under the Euratom safeguards arrangement. In 1998 it signed the Additional Protocol in relation to its safeguards agreements with both the International Atomic Energy Agency (IAEA) and Euratom.

Notes & references

General sources

International Atomic Energy Agency, Country Nuclear Power Profiles: Spain (Updated 2018)